3. Likelihood Maximization Principle

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FS26 MTH345: Deep Generative Models: A Statistical Perspective





Lecture Topics

M1. Foundations of Probabilistic Modelling

1. Fundamentals of Probability Theory
2. Random Variables
3. Likelihood Maximization Principle
4. Basics of Probabilistic Modelling

M2. Likelihood Ratio-based Generative Models

5. Autoregressive Model
6. Energy-based Model
7. Variational Autoencoder
8. Generative Adversarial Network

M3. Generative Models with Other Statistical Distance

9. Integral Probability Metric-based Approaches
10. Wasserstein Distance-based Approaches
11. Score-based Approaches

Lecture Topics

- We have previously formulated basic probability distributions. This lecture reviews how to find optimal distributions using likelihoods.
- Useful materials include:
  - Chapter 8 from Mathematical Statistics and Data Analysis [MSDA] (Rice, 2007)
- Selected examples and text excerpts are reproduced from the materials for educational use only under fair use.

Outline

1. Likelihood

2. Maximum Likelihood Estimation

3. Maximum Likelihood as Divergence Minimization

Motivating Example

- As seen in previous chapters, probability tables (i.e., probability models) describe uncertain real-world outcomes and their probabilities.
- Then, how can we find the probability model that best describes the observed data?
- For instance, consider the following two probability models for a coin-tossing experiment:

[Model A: Fair Coin]

Outcome	H	T
Probability	.5	.5

[Model B: Biased Coin]

Outcome	H	T
Probability	.75	.25

- Suppose we tossed the coin 10 times and observed the following sequence:

Data: H, H, T, H, H, H, H, T, H, H (1)

Which model is more plausible, Model A or Model B?

Motivating Example

- Probability of observing the data under **[Model A: Fair Coin]**:

$$(0.5)^8 \times (0.5)^2 \approx 0.1\% \tag{2}$$

- Probability of observing the data under **[Model B: Biased Coin]**:

$$(0.75)^8 \times (0.25)^2 \approx 0.63\% \tag{3}$$

- A measure of "how likely the observed data is to occur under a specific probability model" is called the *likelihood*.

Motivating Example

- Among all possible probability models we can consider in this setting, which one is optimal in terms of likelihood?

Outcome	H	T
Probability	μ	$1-\mu$

- For a given μ, the likelihood of the observed sequence

$$\text{H, H, T, H, H, H, H, T, H, H} \tag{4}$$

  is given by $\mu^8(1-\mu)^2$. One can prove that the likelihood is maximized at $\hat{\mu} = .8$.
- The principle of preferring a (probability) model yielding a higher likelihood is referred to as the *likelihood maximization*.

Motivating Example





White Heron

designated on Jan. 1, 1995

Q: The city bird of Ulsan Metropolitan City is the white heron. How can we estimate the total population of white herons inhabiting Ulsan?

Image source: Ulsan Metropolitan City (2026).

Motivating Example





White Heron

designated on Jan. 1, 1995

Capture-Recapture Method: First, 10 herons were captured, tagged, and released. Shortly after, 20 herons were recaptured, of which 4 were found to be tagged. What is the total population size of herons in this region?

Motivating Example

- Let n denote the total population size. A simple and intuitive guess can be made using the following proportion:

$$\frac{4}{20} = \frac{10}{n}, \tag{5}$$

  which yields $n = 50$.
- **Q:** Do you agree that $n = 50$ is the most likely true population size?

Motivating Example

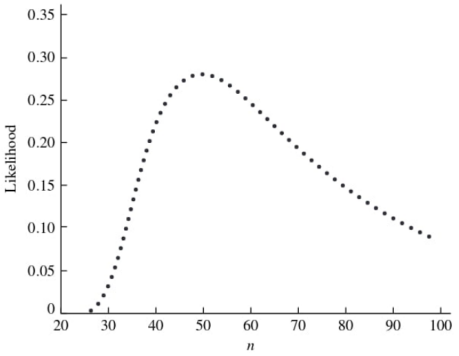


Figure 1.3. [MSDA]: Likelihood plot.

- Given a total population size n, the likelihood L_n is expressed as:

$$L_n = \frac{\binom{10}{4}\binom{n-10}{16}}{\binom{n}{20}}. \tag{6}$$

- As shown in the plot, the likelihood increases with n, attains its maximum at $n = 50$, and decreases thereafter.

Likelihood

- To formulate the likelihood, suppose we observe a sample $x_1, \ldots, x_n$ of size n from a random variable X with PMF p_X (or PDF f_X).
- The likelihood, which is a function of the *data* $(x_1, \ldots, x_n)$ and the *probabilistic model* (p_X or f_X), is given by:

$$L_n = \prod_{i=1}^{n} p_X(x_i). \tag{7}$$

- When the model is parameterized by θ, the likelihood is written as

$$L_n(\theta) = \prod_{i=1}^{n} p_X(x_i; \theta). \tag{8}$$

Likelihood

- For example, in an experiment where a potentially unfair coin is tossed $n = 10$ times, let $X_i \sim \text{Ber}(\mu)$ denote the outcome of the ith toss (with 1 for Heads and 0 for Tails), yielding observations:

$$x_1 = 1, \quad x_2 = 1, \quad x_3 = 0, \quad \ldots, \quad x_{10} = 1.$$

- The underlying probabilistic model is specified by the PMF of $\text{Ber}(\mu)$:

$$p_X(x; \mu) = \mu^x (1 - \mu)^{1-x}, \quad x \in \{0, 1\}, \tag{9}$$

where the parameter of interest is $\theta = \mu \in [0, 1]$.

- Assuming the tosses are independent, the likelihood function is given by:

$$L_{10}(\mu) = \prod_{i=1}^{10} p_X(x_i; \mu) = \mu^{\sum_{i=1}^{10} x_i} (1 - \mu)^{10 - \sum_{i=1}^{10} x_i} = \mu^8 (1 - \mu)^2. \tag{10}$$

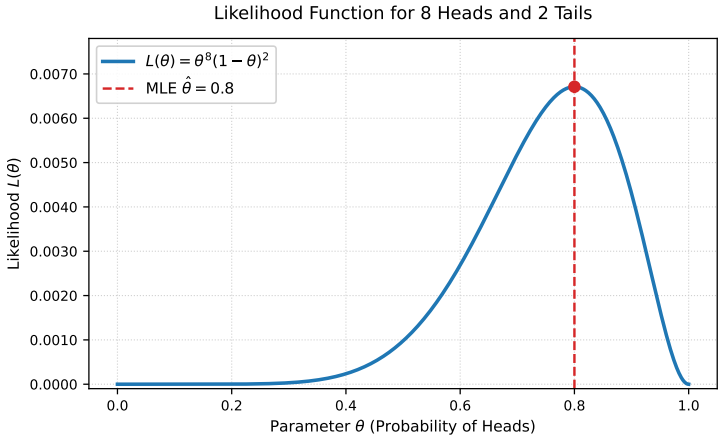
Outline

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2. Maximum Likelihood Estimation

3. Maximum Likelihood as Divergence Minimization

Maximum Likelihood Estimation



- Finding the θ that maximizes the likelihood function is the core of Maximum Likelihood Estimation.

Maximum Likelihood Estimation

Exercise 1

Let $X_1, \ldots, X_n$ be independent and identically distributed (i.i.d.) random variables from $\text{Ber}(\mu)$ (written as $X_i \overset{\text{i.i.d.}}{\sim} \text{Ber}(\mu)$). Find the parameter value(s) $\mu \in [0, 1]$ that maximize the likelihood function.

Maximum Likelihood Estimation

- The maximum likelihood estimate (MLE)[1] is defined as:

$$\hat{\theta}_n^{\text{MLE}} \in \underset{\theta \in \Theta}{\arg\max}\, L_n(\theta) := \left\{ \theta \in \Theta : L_n(\theta) \geq L_n(\theta') \text{ for all } \theta' \in \Theta \right\}, \tag{11}$$

  where Θ denotes the parameter space.

- The logarithm of $L_n(\theta)$ is referred to as the *log-likelihood*:

$$\ell_n(\theta) = \log L_n(\theta), \tag{12}$$

  provided that $L_n(\theta) > 0$.

[1] The acronym MLE is used interchangeably to refer to the maximum likelihood estimate, the maximum likelihood estimator, and maximum likelihood estimation.

MLE: Binomial Distribution

Exercise 2

Let $X_1, \ldots, X_n \overset{\text{i.i.d.}}{\sim} \text{Bin}(m, \mu)$, where the number of trials $m \in \mathbb{N}$ is known and $\mu \in [0, 1]$ is unknown. Find the MLE of μ.

MLE: Binomial Distribution



MLE: Multinomial Distribution

- Let $\boldsymbol{X}_1, \ldots, \boldsymbol{X}_n \overset{\text{i.i.d.}}{\sim} \text{Multin}(1, \boldsymbol{\mu})$, where $\boldsymbol{\mu} = (\mu_1, \ldots, \mu_K)^\top$ lies on the probability simplex. Let's derive the MLE of $\boldsymbol{\mu}$.
- The log-likelihood function is given by

$$l_n(\boldsymbol{\mu}) = \log \left(\prod_{i=1}^{n} \prod_{k=1}^{K} \mu_k^{x_{ik}} \right) = \sum_{k=1}^{K} \left(\sum_{i=1}^{n} x_{ik} \right) \log \mu_k. \tag{13}$$

- Incorporating the simplex constraints, we formulate the optimization problem as

$$\begin{aligned} \max_{\boldsymbol{\mu}} \quad & l_n(\boldsymbol{\mu}) = \sum_{k=1}^{K} \left(\sum_{i=1}^{n} x_{ik} \right) \log \mu_k \\ \text{s.t.} \quad & \sum_{k=1}^{K} \mu_k = 1, \\ & \mu_k \geq 0, \quad k = 1, \ldots, K. \end{aligned} \tag{14}$$

MLE: Multinomial Distribution

Exercise 3

Let $\boldsymbol{X}_1, \ldots, \boldsymbol{X}_n \overset{\text{i.i.d.}}{\sim} \text{Multin}(1, \boldsymbol{\mu})$, where $\boldsymbol{\mu} = (\mu_1, \ldots, \mu_K)^\top$ lies on the probability simplex. Consider the Lagrangian function

$$\mathcal{L}(\boldsymbol{\mu}, \beta) = l_n(\boldsymbol{\mu}) - \beta \left(\sum_{k=1}^{K} \mu_k - 1 \right). \tag{15}$$

1. Find all stationary points of $\mathcal{L}(\boldsymbol{\mu}, \beta)$, and show that they satisfy $\mu_k \geq 0$ for all k.
2. Show that $l_n(\boldsymbol{\mu})$ is concave.

MLE: Multinomial Distribution

MLE: Multinomial Distribution

MLE: Uniform Distribution

Exercise 4

Let $X_1, \ldots, X_n \overset{\text{i.i.d.}}{\sim} \text{Unif}(a, b)$. Find the MLE of $\theta = (a, b)^\top$.

MLE: Gaussian Distribution

Exercise 5

Let $X_1, \ldots, X_n \overset{\text{i.i.d.}}{\sim} \mathcal{N}(\mu, \sigma^2)$. Find the MLE of $\theta = (\mu, \sigma^2)^\top$.

Asymptotic Properties

- Let θ_0 be the true (unknown) parameter value.
- Note that, in the Bernoulli case, the MLE $\hat{\theta}_n^{\text{MLE}} = \bar{x}$ enjoys the following properties:
  1. (Consistency) By the Law of Large Numbers, $\hat{\theta}_n^{\text{MLE}}$ converges in probability to θ_0:

$$P_{\theta_0}\left(|\hat{\theta}_n^{\text{MLE}} - \theta_0| > \epsilon\right) \to 0 \quad \text{as } n \to \infty. \tag{16}$$

  2. (Asymptotic Normality) By the Central Limit Theorem, for any z:

$$P_{\theta_0}\left(\frac{\hat{\theta}_n^{\text{MLE}} - \theta_0}{\sqrt{\text{Var}[\hat{\theta}_n^{\text{MLE}}]}} \leq z\right) \to \Phi(z) := P(Z \leq z) \quad \text{where } Z \sim N(0,1). \tag{17}$$

- These properties hold in general for MLEs under standard regularity conditions.

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3. Maximum Likelihood as Divergence Minimization

Kullback-Leibler Divergence

- The MLE can be naturally interpreted as the minimizer of a *statistical distance* (or *divergence*) between the true data distribution and the parametric model family.
- We first introduce the *Kullback-Leibler divergence* (also called *relative entropy*).
- **Kullback-Leibler Divergence**: Let f_1 and f_2 be PDFs s.t. $\text{supp}(f_1) \subseteq \text{supp}(f_2)$. The *Kullback-Leibler (KL) divergence* from f_2 to f_1 is defined as:

$$\mathcal{D}_{\text{KL}}(f_1 \,\|\, f_2) = \int_{\mathbb{R}^d} f_1(\boldsymbol{x}) \log \left(\frac{f_1(\boldsymbol{x})}{f_2(\boldsymbol{x})} \right) d\boldsymbol{x}. \tag{18}$$

For the discrete case, replacing the integral with a summation over PMFs $p_1(\boldsymbol{x})$ and $p_2(\boldsymbol{x})$ yields the corresponding divergence.[2]

[2] More generally, the KL divergence is defined for probability measures $P \ll Q$ via the Radon–Nikodym derivative $\frac{dP}{dQ}$.

Kullback-Leibler Divergence

Exercise 6

Show that $\mathcal{D}_{\mathrm{KL}}(f_1||f_2) \geq 0$, and that the equality holds if and only if $f_1 = f_2$.

Kullback-Leibler Divergence

- Consider an experiment of tossing a fair coin $n = 10$ times. The true probability of getting heads is $\mu_0 = .5$, identifying the following distribution:[3]

[True Underlying Distribution $p(x; \mu_0)$]

Outcome	1 (Head)	0 (Tail)
Probability	.5	.5

- Given observations of 6 heads and 4 tails from $n = 10$ tosses, the empirical distribution yields the MLE $\hat{\mu}_n = .6$:

[Model Distribution with MLE $p(x; \hat{\mu}_n)$]

Outcome	1 (Head)	0 (Tail)
Probability	.6	.4

[3] Note that, in practice, the true distribution (or parameter) is generally unknown.

Kullback-Leibler Divergence

- The KL divergence between two Bernoulli distributions is given by:

$$\mathcal{D}_{\mathrm{KL}}(\mathrm{Ber}(\mu_1) \,\|\, \mathrm{Ber}(\mu_2)) = \mu_1 \log\left(\frac{\mu_1}{\mu_2}\right) + (1-\mu_1)\log\left(\frac{1-\mu_1}{1-\mu_2}\right). \tag{19}$$

- Applying this formula, $\mathcal{D}_{\mathrm{KL}}(p(x;\mu_0) \,\|\, p(x;\hat{\mu}_n)) = \mathcal{D}_{\mathrm{KL}}(\mathrm{Ber}(0.5) \,\|\, \mathrm{Ber}(0.6))$ evaluates to:

$$\mathcal{D}_{\mathrm{KL}}(\mathrm{Ber}(0.5) \,\|\, \mathrm{Ber}(0.6)) = 0.5 \log\left(\frac{0.5}{0.6}\right) + 0.5 \log\left(\frac{0.5}{0.4}\right) \approx 0.0204. \tag{20}$$

- In contrast, $\mathcal{D}_{\mathrm{KL}}(p(x;\hat{\mu}_n) \,\|\, p(x;\mu_0)) = \mathcal{D}_{\mathrm{KL}}(\mathrm{Ber}(0.6) \,\|\, \mathrm{Ber}(0.5))$ evaluates to:

$$\mathcal{D}_{\mathrm{KL}}(\mathrm{Ber}(0.6) \,\|\, \mathrm{Ber}(0.5)) = 0.6 \log\left(\frac{0.6}{0.5}\right) + 0.4 \log\left(\frac{0.4}{0.5}\right) \approx 0.0201. \tag{21}$$

- Note that $\mathcal{D}_{\mathrm{KL}}(\mathrm{Ber}(0.5) \,\|\, \mathrm{Ber}(0.6)) \neq \mathcal{D}_{\mathrm{KL}}(\mathrm{Ber}(0.6) \,\|\, \mathrm{Ber}(0.5))$, illustrating that the KL divergence is generally asymmetric.

Kullback-Leibler Divergence

Exercise 7

For a given μ_0, show that both $\mathcal{D}_{\mathrm{KL}}(\mathrm{Ber}(\mu_0)||\mathrm{Ber}(\mu))$ and $\mathcal{D}_{\mathrm{KL}}(\mathrm{Ber}(\mu)||\mathrm{Ber}(\mu_0))$ strictly decrease when $\mu < \mu_0$ and strictly increase when $\mu > \mu_0$.

Kullback-Leibler Divergence

Exercise 8

Show that:

$$\mathcal{D}_{\mathrm{KL}}\left(\mathcal{N}(\mu_1, \sigma_1^2) \,\|\, \mathcal{N}(\mu_2, \sigma_2^2)\right) = \frac{(\mu_1 - \mu_2)^2}{2\sigma_2^2} + \frac{1}{2}\left(-1 + \frac{\sigma_1^2}{\sigma_2^2} - \log\frac{\sigma_1^2}{\sigma_2^2}\right). \tag{22}$$

Kullback-Leibler Divergence



Kullback-Leibler Divergence

Exercise 9

Let $p(\boldsymbol{x}) = \prod_{j=1}^{d} f_j(x_j)$ and $q(\boldsymbol{x}) = \prod_{j=1}^{d} g_j(x_j)$. Show that:

$$\mathcal{D}_{\mathrm{KL}}\left(\prod_{j=1}^{d} f_j(x_j) \,\middle\|\, \prod_{j=1}^{d} g_j(x_j)\right) = \sum_{j=1}^{d} \mathcal{D}_{\mathrm{KL}}\left(f_j(x_j) \,\|\, g_j(x_j)\right). \tag{23}$$

Kullback-Leibler Divergence

- For multivariate Gaussians with diagonal covariance matrices, the KL divergence is given by:

$$\begin{aligned}&\mathcal{D}_{\mathrm{KL}}\left(\mathcal{N}(\boldsymbol{\mu}_1, \mathrm{diag}(\boldsymbol{\sigma}_1^2)) \,\|\, \mathcal{N}(\boldsymbol{\mu}_2, \mathrm{diag}(\boldsymbol{\sigma}_2^2))\right) \\ &= \frac{1}{2}\sum_{j=1}^{d}\left(\frac{(\mu_{1,j}-\mu_{2,j})^2}{\sigma_{2,j}^2} - 1 + \frac{\sigma_{1,j}^2}{\sigma_{2,j}^2} - \log\frac{\sigma_{1,j}^2}{\sigma_{2,j}^2}\right).\end{aligned} \tag{24}$$

- For general multivariate Gaussians, the divergence is:

$$\begin{aligned}&\mathcal{D}_{\mathrm{KL}}\left(\mathcal{N}(\boldsymbol{\mu}_1, \Sigma_1) \,\|\, \mathcal{N}(\boldsymbol{\mu}_2, \Sigma_2)\right) \\ &= \frac{1}{2}\left(\mathrm{tr}(\Sigma_2^{-1}\Sigma_1) + (\boldsymbol{\mu}_1-\boldsymbol{\mu}_2)^\top\Sigma_2^{-1}(\boldsymbol{\mu}_1-\boldsymbol{\mu}_2) - d - \log\frac{|\Sigma_1|}{|\Sigma_2|}\right).\end{aligned} \tag{25}$$

- For proof details, see `https://statproofbook.github.io/P/mvn-kl.html`.

Beyond the KL-Divergence

- In comparing $\mathrm{Ber}(\mu_1)$ and $\mathrm{Ber}(\mu_2)$, how about simply using $|\mu_1 - \mu_2|$?
- **Total Variation Distance**: Let f_1 and f_2 be PDFs. The *total variation (TV) distance* between them is defined as:

$$\delta_{\mathrm{TV}}(f_1, f_2) = \frac{1}{2} \int_{\mathbb{R}^d} |f_1(\boldsymbol{x}) - f_2(\boldsymbol{x})| \, d\boldsymbol{x}. \tag{26}$$

  For discrete distributions with PMFs p_1 and p_2, replacing the integral with a sum yields:

$$\delta_{\mathrm{TV}}(p_1, p_2) = \frac{1}{2} \sum_{\boldsymbol{x}} |p_1(\boldsymbol{x}) - p_2(\boldsymbol{x})|. \tag{27}$$

- In particular, for two Bernoulli distributions, $\delta_{\mathrm{TV}}(\mathrm{Ber}(\mu_1), \mathrm{Ber}(\mu_2)) = |\mu_1 - \mu_2|$.

Beyond the KL-Divergence

- Note that

$$\begin{aligned}\delta_{\mathrm{TV}}(f_1, f_2) &= \frac{1}{2}\int_{\mathbb{R}^d} |f_1(\boldsymbol{x}) - f_2(\boldsymbol{x})| d\boldsymbol{x} \\ &= \frac{1}{2}\int_{\mathbb{R}^d} \left| \frac{f_1(\boldsymbol{x})}{f_2(\boldsymbol{x})} - 1 \right| f_2(\boldsymbol{x}) d\boldsymbol{x} \\ &= \frac{1}{2}\int_{\mathbb{R}^d} g\left(\frac{f_1(\boldsymbol{x})}{f_2(\boldsymbol{x})}\right) f_2(\boldsymbol{x}) d\boldsymbol{x}\end{aligned} \tag{28}$$

  where $g(t) = |t-1|$ is a strictly convex function with $g(1) = 0$.

- Similarly,

$$\mathcal{D}_{\mathrm{KL}}(f_1 || f_2) = \int_{\mathbb{R}^d} g\left(\frac{f_1(\boldsymbol{x})}{f_2(\boldsymbol{x})}\right) f_2(\boldsymbol{x}) d\boldsymbol{x} \tag{29}$$

  where $g(t) = t \log t$ is also a strictly convex function with $g(1) = 0$.

Beyond the KL-Divergence

- **f-Divergence (Csiszár, 1967)**: Let f_1 and f_2 be PDFs s.t. $\text{supp}(f_1) \subseteq \text{supp}(f_2)$. For a given convex function $f : \mathbb{R}^+ \to \mathbb{R}$ satisfying $f(1) = 0$, the *f-divergence* of f_1 w.r.t. f_2 is defined as:

$$\mathcal{D}_f(f_1 \,\|\, f_2) = \int_{\mathbb{R}^d} f\left(\frac{f_1(\boldsymbol{x})}{f_2(\boldsymbol{x})}\right) f_2(\boldsymbol{x})\, d\boldsymbol{x}. \tag{30}$$

- Notable examples include the KL divergence, total variation (TV) distance, α-divergence (Rényi, 1961), and Jensen-Shannon (JS) divergence:[4]

$$\mathcal{D}_{\text{JS}}(f_1 \,\|\, f_2) = \frac{1}{2}\mathcal{D}_{\text{KL}}\left(f_1 \,\middle\|\, \frac{f_1 + f_2}{2}\right) + \frac{1}{2}\mathcal{D}_{\text{KL}}\left(f_2 \,\middle\|\, \frac{f_1 + f_2}{2}\right). \tag{31}$$

[4] Although $\mathcal{D}_{\text{JS}}(f_1 \,\|\, f_2)$ is not a metric, its square root $\sqrt{\mathcal{D}_{\text{JS}}(f_1 \,\|\, f_2)}$ is a valid metric (Endres and Schindelin, 2003).

Beyond the KL Divergence

Exercise 10

Show that $\mathcal{D}_f(f_1 \,\|\, f_2) \geq 0$. Furthermore, show that when f is non-linear, equality holds if and only if $f_1 = f_2$. In this problem, you may use Jensen's inequality without proof: for any convex function ψ,

$$\psi(\mathsf{E}[X]) \leq \mathsf{E}[\psi(X)],$$

where equality holds if and only if X is constant or ψ is linear.

Beyond the KL Divergence

Exercise 11

Show that the Jensen-Shannon divergence is an f-divergence. Specifically, verify that

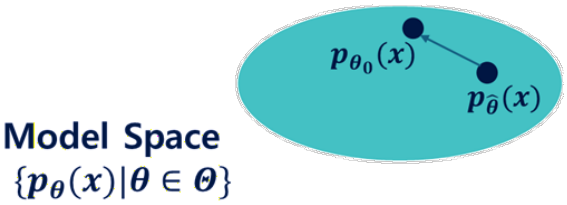
$$f(t) = \frac{1}{2}\left(t \log t - (t+1)\log\left(\frac{t+1}{2}\right)\right) \tag{32}$$

satisfies $f(1) = 0$ and is strictly convex on $\mathbb{R}^+$, and show that

$$\mathcal{D}_{\mathrm{JS}}(f_1 \,\|\, f_2) = \int_{\mathbb{R}^d} f_2(\boldsymbol{x}) f\left(\frac{f_1(\boldsymbol{x})}{f_2(\boldsymbol{x})}\right) d\boldsymbol{x}. \tag{33}$$

Beyond the KL-Divergence

MLE as a Divergence Minimizer



- Recall that MLE satisfies the consistency, i.e., $\hat{\theta}_n^{\text{MLE}}$ converges to θ_0 as $n \to \infty$.
- The consistency implies $\lim_{n\to\infty} \mathcal{D}_{\text{KL}}(p(x;\theta_0)||p(x;\hat{\theta}_n)) = 0$.
- **Q:** What if the true data distribution does not belong to $\{p(\cdot;\theta) : \theta \in \Theta\}$? For example, in the ancient Korean die (Juryeonggu) problem, we might assume that all square faces are equally likely, but the true distribution may not be.

MLE as a Divergence Minimizer

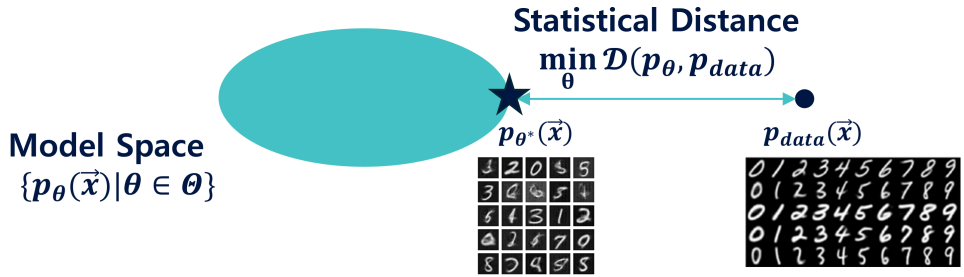
Exercise 12

Let f_{data} denote the true data distribution generating observations $x_1, \ldots, x_n$. The empirical KL-divergence is defined as:

$$\widehat{\mathcal{D}}_{\text{KL}}(f_{\text{data}} \,\|\, f_\theta) = \frac{1}{n} \sum_{i=1}^{n} \log \frac{f_{\text{data}}(x_i)}{f_\theta(x_i)}. \tag{34}$$

Show that the MLE $\hat{\theta}_n^{\text{MLE}}$ is a minimizer of the empirical KL-divergence.

MLE as a Divergence Minimizer



- That is, maximizing the (log-)likelihood can be interpreted as minimizing the (empirical) KL-divergence.

Handwritten digit images are from Wolterink et al. (2017).

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